

JOHN TURNAGE

Department of Mathematics & Scientific Computing and Imaging Institute

University of Utah

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Education

University of Utah

PhD, Mathematics

2022 – present

Expected May 2027

- Advisor: Akil Narayan
- Dissertation (working title): *Optimal Sampling and Weighted Least Squares for Operator Learning, Quadrature, and Uncertainty Quantification*

Wake Forest University

MS, Mathematics

2020 – 2022

- Advisor: John Gemmer
- Thesis: *Geometric Mechanics and Control*

University of Georgia

BS, Physics & BS, Mathematics; Minor, Philosophy

2014 – 2019

Research Interests

Approximation theory and probability for scientific computing: learning functions, operators, and probability distributions from finite data, with quantitative guarantees of stability, accuracy, and uncertainty.

Operator learning · optimal sampling and weighted least squares · uncertainty quantification · function-space inverse problems · quadrature · model order reduction · measure transport and generative models · structure-preserving approximation

Research Experience and Training

John Tukey Summer Graduate School: Mathematics of Generative Models

Summer 2026

Simons Laufer Mathematical Sciences Institute (SLMath, formerly MSRI); two weeks

Graduate Research Intern, Sandia National Laboratories

Summer 2025

Optimization and Uncertainty Quantification Group, Center for Computing Research

Mentor: John D. Jakeman

- Benchmarked transport-based generative models (score-based diffusion, flow matching, normalizing flows, GANs) on high-dimensional goal-oriented inverse problems, in which a low-dimensional quantity of interest is conditioned directly on data [5].

Publications and Preprints

- [1] **J. Turnage**, J. D. Jakeman, and A. Narayan. *Gaussian Residual Inference for Operator Learning from Full-Field Observations*. In preparation, 2026.
- [2] F. Bělík[†], A. Narayan[†], and **J. Turnage**[†]. *Constructive Tchakaloff Results and Well-Conditioned Quadrature through Randomized Least Squares*. Submitted, 2026. [arXiv:2609.09506](#).
- [3] M. Lowery, **J. Turnage**, Z. Morrow, J. D. Jakeman, A. Narayan, S. Zhe, and V. Shankar. *Kernel Neural Operators (KNOs) for Scalable, Memory-efficient, Geometrically-flexible Operator Learning*. Transactions on Machine Learning Research, 2026. [arXiv:2407.00809](#).
- [4] **J. Turnage**, M. Lowery, J. D. Jakeman, Z. Morrow, A. Narayan, and V. Shankar. *An Optimal Weighted Least-Squares Method for Operator Learning*. Submitted, 2025. [arXiv:2512.11168](#).
- [5] **J. Turnage**[†], M. Lowery[†], and J. D. Jakeman. *Benchmarking Generative Models for Solving High-Dimensional Goal-Oriented Inverse Problems*. Computer Science Research Institute Summer Proceedings 2025, Sandia National Laboratories, pp. 286–303, [SAND2025-14267O](#), 2025.

[†]Equal contribution.

Honors and Awards

Outstanding Graduate Student Award	2026
<i>Department of Mathematics, University of Utah: awarded to graduate students for excellence in teaching and research</i>	
Graduate Student Travel Award	2026
<i>Department of Mathematics, University of Utah</i>	
Graduate Student Professional Development Supplement Award	2026
<i>Graduate School, University of Utah</i>	
NSF RTG Fellowship in Optimization and Inversion	2024 & 2026
<i>Department of Mathematics, University of Utah: semester (2024) and summer (2026) research funding</i>	
Outstanding Student Award	2021
<i>Department of Mathematics, Wake Forest University: awarded to the two top graduating master's students</i>	
Undergraduate Research Award	2017
<i>Department of Physics, University of Georgia: for outstanding undergraduate physics research</i>	
Ted L. Simmons Memorial Award	2017
<i>Department of Physics, University of Georgia: for outstanding achievement by a junior physics student</i>	

Conference and Seminar Presentations

Invited Minisymposium Talks

<i>Christoffel-Weighted Approximation for Stable Operator Learning with Posterior Uncertainty.</i> SIAM Conference on Computational Science and Engineering (upcoming).	February 2027
<i>Certifiable Operator Learning via Optimally Weighted Least Squares.</i> SIAM Conference on Mathematics of Data Science (upcoming).	November 2026
<i>An Optimal Weighted Least Squares Framework for Operator Learning.</i> WCCM-ECCOMAS.	July 2026
<i>An Optimal Weighted Least Squares Framework for Operator Learning.</i> U.S. National Congress on Computational Mechanics.	July 2025

Contributed and Seminar Talks

<i>Uncertainty Quantification in Operator Learning.</i> Wasatch SIAM Student Chapters Conference.	April 2026
<i>Uncertainty Quantification in Operator Learning.</i> Applied Math Collective Seminar, University of Utah.	April 2026
<i>Certifiable Operator Learning.</i> FASTMath Institute, Uncertainty Quantification Team (virtual).	July 2025
<i>Operator Learning.</i> Applied Math Collective Seminar, University of Utah.	November 2024
<i>Dynamics of a Rolling Disk.</i> Integrating Research in Science, Wake Forest University.	April 2022

Posters

<i>An Optimal Weighted Least Squares Framework for Operator Learning.</i> SIAM Conference on Mathematics of Data Science.	October 2024
<i>An Optimal Weighted Least Squares Framework for Operator Learning.</i> NSF Computational Mathematics PI Meeting.	July 2024

Teaching Experience

Course Title	Term	Students	Position
Quantitative Reasoning	Fall 2023	61	Instructor of record
Differential Equations & Linear Algebra	Summer 2023	40	Instructor of record
Differential Equations & Linear Algebra	Spring 2023	48 (2 sections)	Lab TA
Engineering Calculus I	Fall 2022	65 (2 sections)	Lab TA

All courses at the University of Utah.

Committees

College of Science Scholarships and Awards Committee

2026

College of Science, University of Utah

- One of three graduate students on this faculty committee; reviewed undergraduate applications and selected recipients of College of Science scholarships and awards.

Graduate Student Advisory Committee

2023 – 2025

Department of Mathematics, University of Utah

- Hiring Chair (2023–24): reviewed applications for faculty positions and made recommendations to the department's hiring committee on behalf of graduate students.
- Retention, Promotion, and Tenure Committee Member (2024–25): reviewed faculty retention, promotion, and tenure cases and made recommendations to the department on behalf of graduate students.

Organizing

Wasatch SIAM Student Chapters Conference

April 2026

Co-organizer

- Annual regional student conference hosted in rotation by the University of Utah, BYU, and Utah State University; about 30 attendees and 15 speakers.

SIAM Student Chapter

2024 – present

Executive Officer and Webmaster, University of Utah

- Help organize chapter activities, including career talks by invited speakers and chapter entries in the COMAP modeling competitions; maintain the chapter website.
- Developing a monthly hands-on workshop on data-driven modeling.

SIAM Conference on Computational Science and Engineering

March 2025

Minisymposium co-organizer: Accelerating Inference, UQ, and Optimization with Machine Learning Models

Applied Math Collective

2024 – 2025

Organizer, University of Utah

- Scheduled a weekly graduate student-led applied mathematics seminar and recruited speakers from within the department.

References

Akil Narayan

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John D. Jakeman

Internship mentor
Principal Member of Technical Staff
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